

# Evaluating the Performance of Hierarchical Clustering algorithms to Detect Spatio-Temporal Crime Hot-Spots

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**Abstract**—The constant growth in urbanization is a cause of significant social and economical transformations in urban areas. Areas where crime rates are above the normal level, are known as crime hot-spots. The increase in urban population is posing challenges related to the management, services and safety from criminal activities. It is important to keep an eye on criminal activities and for the law enforcement agencies, being able to provide much needed safety of public is an increasingly complex task. This complex task can be handled by new technologies which can help these agencies to effectively analyze and understand the different crime trends and patterns with respect to their geographic locations. This paper uses Hierarchical Density-based spatial clustering of applications with noise (HDBSCAN) to find spatio-temporal crime hot-spots by clustering and the results shows that this technique outperforms others.

**Index Terms**—Hot-spot detection, Clustering, Hierarchical Clustering, Agglomerative Clustering, HDBSCAN

## I. INTRODUCTION

Among other different common social problems, crimes severely affects the living standard of life and derail social and economical development of a society [1]. While planning to move to a new city or traveling to a new location, crimes rate of that location is considered an important factor in decision making [2]. Due to gradual increase in crime rates, the need of hour of law enforcement agencies is to improve crime analytic by using new data mining approaches, hence providing better protected and safer communities [3].

Because of the constant growth in urbanization, 21<sup>st</sup> Century can be referred as the "Century of City", it shows that gradual

migration into urban areas is underway [4]. Among other challenges being faced due to growth in urbanization, causing social problems and compromising public safety, health, child development and education, crime spiking is becoming one of the most important social problems [5] [6]. Fig. 1 shows the crime spikes over the years in the city of New York [7].

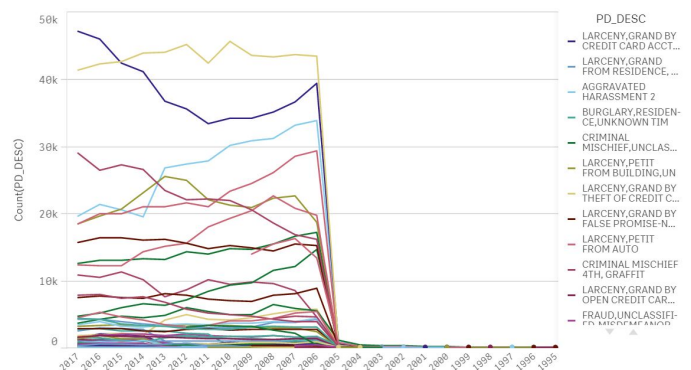


Fig. 1. Crime Spikes over the years in NYC

In order to over come the crime spiking, effective policies needs to be in place. For this purpose, predictive policing is the one of the first which uses analytical techniques either to identify the current crime dense areas or to find out the likely places where crimes could occur [8]. Using statistical techniques, a crime scene can lead us to the type of crime, perpetrator, victim and location of that crime in future. Higher crime rates increased complexities, which led to the new

technologies helping police to analyze and understand crime trends and patterns [9]. It is well known in this context that criminal incidents are not distributed uniformly within a city.

It can be understood that crime can be a location-specific characteristic, and some locations may possess higher crime risk than others [10]. Hence, crime is not distributed randomly or uniformly within a city [11]. In this regard, mapping of crime hot-spots can help understand the reasons behind the frequent occurrence of crimes in those areas. This is a crucial factor for both citizens and law enforcement agencies. Because for the former, it is an important decision to move to a certain place and settling there, and for the latter, effective policies needs to be in place to be able to control and counter the crimes.

Different types of crimes and overall perception of the safety of citizens in a city are major factors that directly plays a role in quality of the lives of residents. Certain types of criminal incidents such as larceny, identity theft or even pick-pocketing can be a cause of distress in a person's life and affects his well-being. Criminology develops and studies different theories regarding criminal behavior from different perspectives to address these issues. It is understood from [7] that frequency of different types of criminal incidents are not distributed uniformly. Different types of crimes occurred in NYC over the span of more than 10 years, with their frequency counts are show in the following figure.

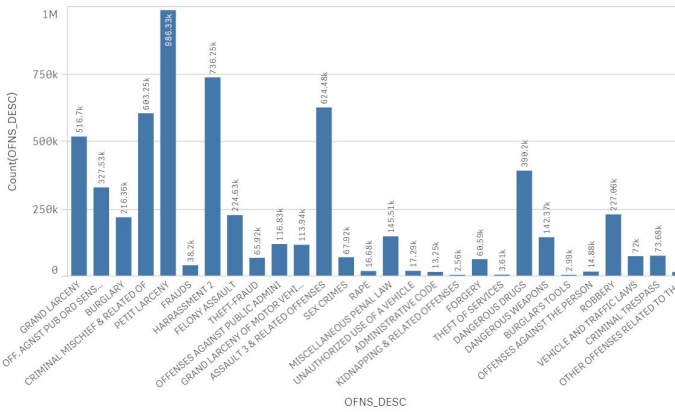


Fig. 2. Frequency distribution of different criminal events in NYC

Hot-spot mapping has emerged as an analytical technique for identifying places with higher risk to appropriately allocate police resources and to perform effective interventions [12] [13]. An experimental evaluation of relevant methods, including point mapping, grid thematic mapping, standard deviation spatial ellipses, kernel density estimation and thematic mapping of administrative units has been presented [12].

Considering the threats and challenges imposed by crimes, this study aims to find the crime hot-spots using DBSCAN and HDBSCAN and compares the results of both techniques in terms of seconds taken to make the clusters on the dataset of 100k instances. The dataset being used is provided by the Gov. of the City of New York [7].

The rest of paper is organized as follows. Section II discusses the Literature Review, Methodology and Results are mentioned in section III and IV respectively. The findings are concluded in section V.

## II. LITERATURE REVIEW

Prevention of crimes have always been an important problem for everyone in the society. It has gained a lot attraction not only from government, but from academia as well and both have been on their toes to provide most effective and viable solutions. Extensive studies of criminal justice depicts that the distribution of criminal events within a city are not equal [9]. Crime rates can vary according to geographic location, some areas can be identified as low-risk and high-risk areas. The trends of crimes can change with seasonal patterns and time of the year as well.

Considering crime a location-specific characteristic, analyzing the crime with respect to location, has grown in last decade. Among other approaches, hot-spot analysis is considerable important and popular approach [14] [15] [16] [17]. For this purpose, point pattern analysis [18], [19] and clustering [20] is commonly used. Another method is the discovery of trends and patterns using some knowledge discovery and data mining techniques [21], such as association rule mining [22], text mining and spatial analysis [18], and self-organizing maps [23]. It is found from the studies that in some locations, the perception of crime is, in fact, higher than the actual risk level [24].

[25] worked on thefts and robberies data of Shanghai, 2009, to identify hot-spots using ArcGIS9.3. By analyzing the hot spots detected from the data of theft and robbery, it was found that central city edge zone, suburban areas and central city core area of the city is a crime cold spot in most of the seasons. [26] performed comparative analysis using number of techniques including the Besag and Newell statistic, Geographical Analysis Machine, and Killdorff's spatial scan statistics. It was found out that the size and locations of the detected clusters are sensitive to the chosen parameters of each method by working on criminal. In [27], Hot Spots prediction model based on mixed spatial-temporal characteristics was used on the data of main city zone of Nanchang ranging from 2014 to 2015. It was found that optimal performance can be achieved by the prediction model if crime statistics are conducted on weekly basis.

[28] used number of techniques i.e. Spatio-Temporal Neural Network (STNN), Decision Tree, Guassian Naive Bayes, Random Forest, K-nearest neighbors, Logistic Regression, Multi-layer perception. It was found that STNN outperformed others with 81.50% accuracy. These techniques were applied on call-for-service data provided by the Portland, Oregon Police Bureau (PPB) for a 5-year periof from March 2012 through the end of December 2016. Similarly, [29] used Random Forest on Data of 12 years, 2003 to 2015, San Francisco (US), of crime records and on one from Natal (Brazil) with 10 years (2006-2016) of crime records. As per the findings, features such as street network contains

important information regarding crime based activities. In [30], Kernel Density Estimation (KDE) was applied on data regarding crimes occurred in Manila, Philippines from the year 2012 to 2016. From the results, it was concluded that criminal activities in Manila are at peak around 8:00 PM to 4:00 AM.

Authors of [31] used spatio-temporal kernel density estimation (STKDE) on the data of residential burglaries in Baton Rouge, Louisiana in 2011. It was found that Southwest area of Baton Rouge is the high-risk area. Similarly, spatio-temporal ordinary kriging was used in [32], on the data set of crimes occurred in Philadelphia from January 2011 to December 2016 and it was concluded that the technique secured 90.52% sensitivity.

[9] used DBSCAN on the crimes dataset of New York city and Chicago and achieved significant results. [33] secured 72% accuracy by detecting the hot-spots on data from three different geographical areas from Sweden i.e. Stockholm, Gothenburg and Malmo. They used Kernel Density Estimation (KDE) Spatial Mapping Technique to do so. Similarly, [34] applied Network based KDE on the crime reports of 2012, Faisalabad, Pakistan. After the detection of the hot-spots, it was found that there was a significant decrease in crimes of those areas. [35] applied apriori algorithm on two different real-world crimes datasets for Denver, CO and Los Angeles, CA and found hot-spots.

### III. METHODOLOGY

The dataset used in this study is Crimes Dataset of NYC [7]. It comes from *data.cityofnewyork.us*, a website for open data about different disciplines concerning the city of New York, USA. It is in CSV format, containing instances as crimes reported over the years. The dataset is comprised of 35 attributes. Two of them are Latitude and Longitude which will be used for clustering as they point to the location on which the crime was committed.

The following image depicts the steps performed in this paper to find the hot-spots.



Fig. 3. Methodology

First of all, pre-processing is performed on the data to deal with missing values and outliers. All those data points which didn't represent the locations from NYC, were identified and removed. And to handle the missing values, *FillByMean* method was used, because the outliers were removed, therefore the mean will always point to within the region of interest. Because to perform clustering, there cannot be any null values. To perform clustering, two Hierarchical Clustering algorithms are used. (A) *Hierarchical Agglomerative Clustering (HAC)* and (B) *Hierarchical Density-based spatial clustering of*

*applications with noise (HDBSCAN)*. As both algorithms are implemented in Python, both used *euclidean distance* for distance calculations, while HAC's default number of clusters are 2 but 6 were mentioned and computed with *ward* linkage method because ward minimizes the variance of the clusters being merged. And HDBSCAN's default number of minimum clusters to be computed are 5, we mentioned 6 with 350 minimum samples which is the number of samples in a neighborhood for a point to be considered a core point.

#### A. Hierarchical Agglomerative Clustering

Agglomerative clustering is very common type of hierarchical clustering that is used to group objects in clusters based on their Similarities. It is a bottom-up clustering in which we assign each object to its own cluster. In this technique we join the two most similar clusters between each cluster to compute the similarity and repeat the steps until there is only single cluster left. Before any clustering is performed, a distance function is used to determine the proximity matrix containing the distance between each point. It should be understood that number of clusters that are to be computed, must be mentioned. Following are the steps below to perform agglomerate hierarchical clustering.

- Preparation of data.
- Computation of dis-similarity information between every part of objects in the data set to make each data point to be a cluster
- Repeat: Using Linkage functions to group objects into hierarchical cluster tree that based on the distance information generated at step1. By using that Linkage functions objects and clusters are in close proximity are now Linkage together.
- Determining from where to cut the hierarchical tree into cluster until only single cluster remain.

These steps were performed on the dataset with 100k data points of two aforementioned attributes, using *Google Colab* as it provides up to 25GB of RAM. The algorithm was applied in Python, using the default parameters. HAC has a time complexity of  $O(n^3)$  and it requires  $O(n^2)$  memory. Hence, it could perform the clustering on up to 55k data points, because to perform calculations on data points more than 55k, it requires RAM more than that. The algorithm was performed initially on 10k data with an interval of 10k.

#### B. HDBSCAN

HDBSCAN is a famous clustering algorithm. It is the extended form of DBSCAN by converting it into a hierarchical clustering algorithm and then by using a technique to extract a flat clustering. HDBSCAN is a notebook that provides an overview of how the algorithm works. HDBSCAN is a library that contains the suit of tools to use unsupervised learning to find clusters or a deep region of dataset. To perform the computations, minimum number of clusters Following are the steps for HDBSCAN.

- Transformation of the space according to the density.

- Creating the minimal spanning tree of the distance weighted graph.
- Build a cluster hierarchy of connected components.
- Summarize the cluster hierarchy based on minimum cluster size
- Extraction of stable clusters from the summarized tree.

The algorithm was applied on aforementioned data points.

#### IV. RESULTS

Considering the memory requirement of HAC and its time complexity, the results shows that HDBSCAN performed better in terms of time required to perform clustering. It must be noted that, on the basis of data points, HDBSCAN forms number of clusters, while in HAC, you have to mention K number of clusters. Moreover, HAC could perform calculations on upto 55k data points only consuming more than 107 seconds and occupying almost 23GB RAM, while HDBSCAN performed the calculations on 100k data points in approximately 7 seconds. The results in terms of time required to perform clustering, clearly shows that HDBSCAN outperformed HAC in terms of memory and time requirement on numerical data points. The performance of HAC in terms of seconds required to performed calculations is shown in Fig. 4 below.

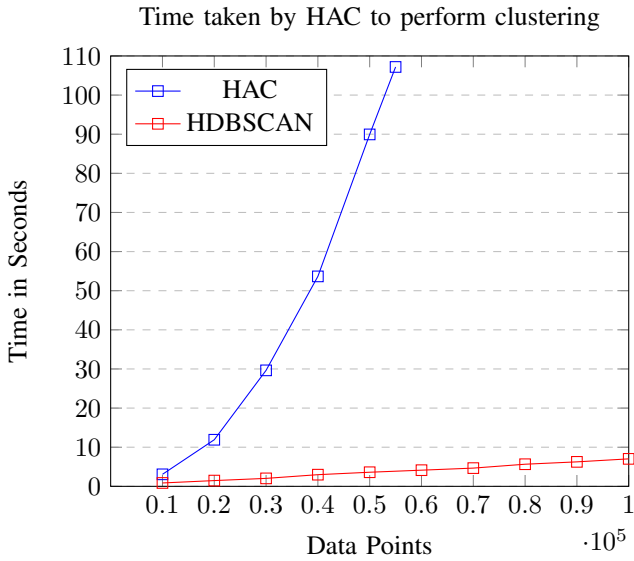


Fig. 4. Performance of HAC

Moreover, Fig. 5 shows the clusters computed by HAC on 55k data points, and Fig. 6 shows the clusters computed by HDBSCAN on 100k data points.

#### V. CONCLUSION

This paper presented a comparative analysis of two hierarchical clustering algorithms to detect crime hot-spots in urban areas. The algorithms were applied on the data set of New York City which resulted in fine-grained information about where criminal events are being occurred frequently and this

information can be used to take suitable measures regarding effective policing and using man power to counter those events. Moreover, the analysis showed that HDBSCAN outperformed HAC in terms of time and memory requirements.

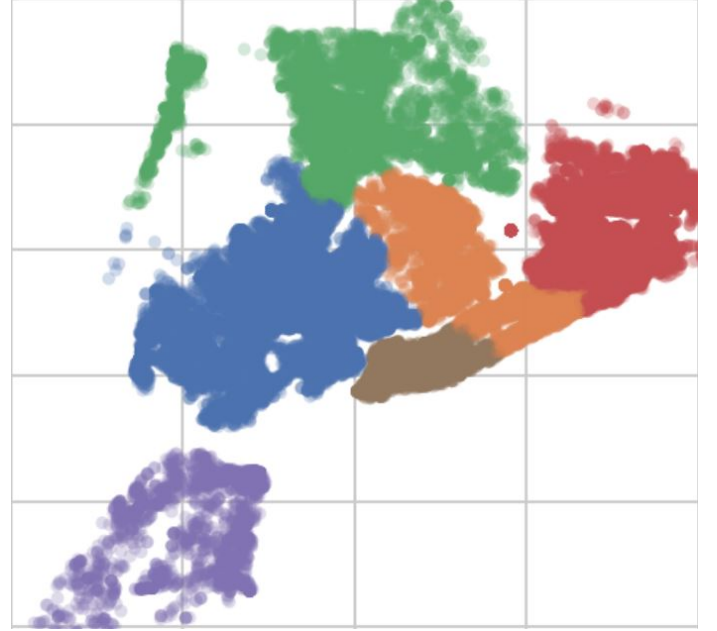


Fig. 5. Clusters computed by HAC on 55k Data Points

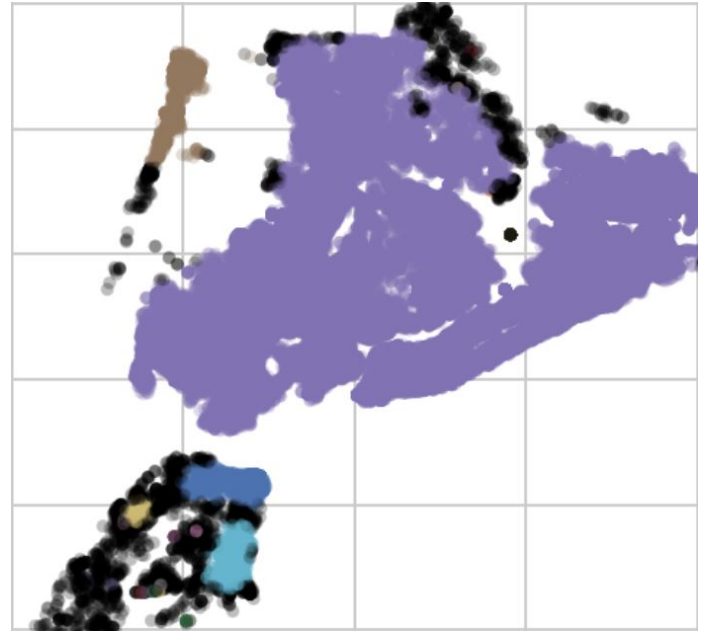


Fig. 6. Clusters computed by HDBSCAN on 100k Data Points

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